



UNIVERSITY OF AMSTERDAM

Digital Advertising & Pricing Models

Personalization - Personalized Targeting

Digital Marketing and Analytics: Week 5

Academic Year 2025-2026 – Semester 1 – Block 2

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Digital Advertising & Pricing Models

Digital Advertising: Pricing Models

No matter which advertising format is used (i.e. SEA, display or social media) – online ads are priced according to one these techniques:

CPV-CPM: Cost Per View or Mille

you pay for each or / every 1,000 views

CPT: Cost Per Time

you pay per time unit, e.g. day

CPC: Cost Per Click

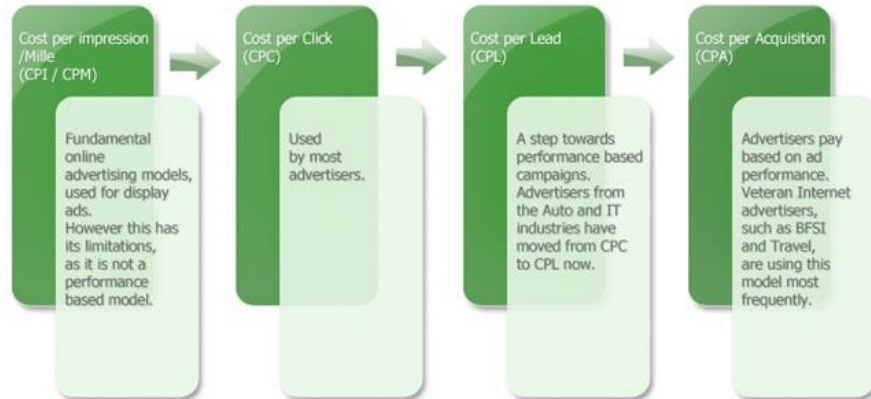
you pay if someone clicks the ad

CPA: Cost Per Acquisition/Conversion

you pay if one clicks the ad & buys your product

RTB: Real Time Bidding

real time auction: mostly based on CPC (can apply to all above)



Digital Advertising: Pricing Models

Cost per Impression (View or Mille)

- Every display is charged: though there is no certainty whether the viewer sees the ad (see: Lecture 1 !)
- Often measured by CPM (cost per thousand impressions)
- Response to the advertisement is not guaranteed – However often the LOWEST cost for advertiser
- From the publisher's perspective, CPM is the best choice because of the predictable revenue and measurable results.

CPM: very similar to CPV to maximize the number of people get to view the product, brand or service.

CPV/CPM are useful for getting more people are aware of the new brand or a specific product. They do not guarantee/monitor ad performance.

This method is considered as least risky for Publisher and most risky for Advertiser – since the advertiser has to pay even if nothing at all happens.



Digital Advertising: Pricing Models

Cost per Actions (Cost-per-Click or Conversions/Acquisitions)

- Displayed for free, pay only when the defined action is taken: this could be Click or Conversion
- Measured by, for example, CPC (cost per click)
- There is a proof that the user has taken the action (compared to CPM/CPV)

Cost per Click: used when an advertiser wants more people to click on their product or brand advertisements. Although Click is a weak measure of campaign/ad performance. However, **CPC is popular because mostly found as nicely balanced between cost and the trackability (by clicks)**

Cost per Conversion/Acquisition: advertiser pays when ad objective is reached: thus purchase, installation the app etc. (conversion objective reached)
This method is considered as least risky for Advertiser and most risky for Publisher – since the advertiser pays only when there is a conversion-acquisition.



Calculating Costs: An Example - CPM, CPC and CPA

Let's say we have an advertisement on the BBC website with:

Cost per 1000 views (CPM): €5 Click Through Rate (CTR): 0.1% and Conversion Rate (CR) : 2%

Cost per sale (in Cost Per Mille):

$\text{CPM}/1000 * 1/\text{CTR} * 1/\text{CR}$ Then > $\text{€}5/1000 * 1/0.001 * 1/0.02 = \text{€}250$



Quality Score: Impacts (too) on CPC

Do you remember the “Quality Score” (used by Google) from Lecture 4, that is used to determine your # position in paid search results in SEA

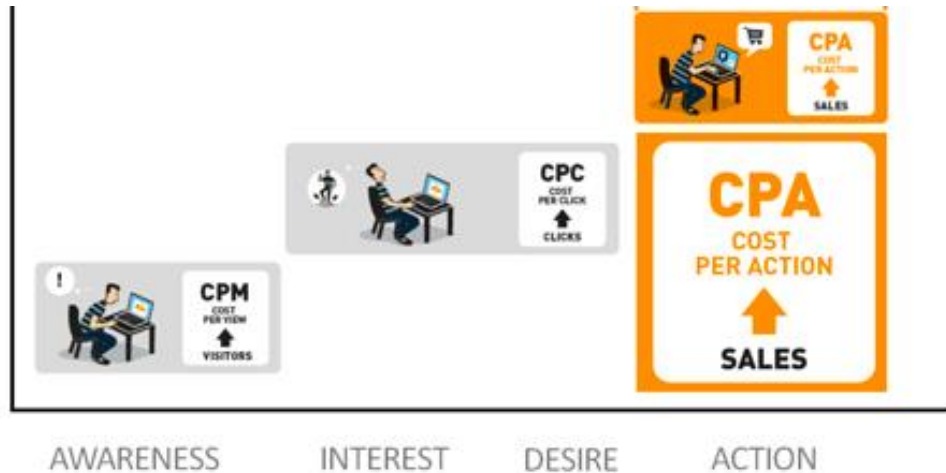
Quality Score also influences your CPC: cost-per-click

Websites with a higher/ **above-average Quality Score** gets a **discount on their CPC** – the opposite applies for those with a below-average QS



Digital Ad Pricing Models: Comparison

- **CPM** is agreeable if the only (and mere) objective is to create **awareness** for new products or campaigns (i.e. Branding)
- **CPC** is often used when the objective is to create/measure creation of an **“interest”** by the ad – and not necessarily selling the product.
- **CPA** is agreeable for a heavily **conversion oriented ad** with relatively (very) high “click/ conversion” ratio



Digital Ad Pricing Models: Comparison

The advantages for Publisher and Advertiser should be assessed for each model for decision making.

For example: In CPA the brand (advertiser) only pays for the concrete results (conversion) – but on the other hand CPA pricing can be really costly.

	COST PER MILLE	COST PER CLICK	COST PER ACTION (CONVERSION)
Publisher			
Advantages	Knows the expected revenue per impression. No concern about CTR	The publisher can block some advertisers	Potentially highest payout.
Disadvantages	High traffic required. Clicks pay nothing.	High CTR required Unpredictable revenue	Unpredictable revenue Clicks pay nothing.
Advertiser			
Advantages	Knows the expected daily impressions and costs.	Indicator for banner quality. Can block some websites to display the campaign.	Pays according to results, performance only. High correlation between ads and sales or leads Indicator of campaign and banner quality.
Disadvantages	Weak correlation between ads and sales or leads. Poor conversion and performance advertising.	Weak correlation between ads and sales or leads. Vulnerable to click frauds.	It can imply higher cost, but also higher ROI.

Real Time Bidding: How (Cost-Per) Campaigns are Priced

Early era of online advertising: there were **fixed price** (being updated manually) for ad placements on web-sites and other online platforms

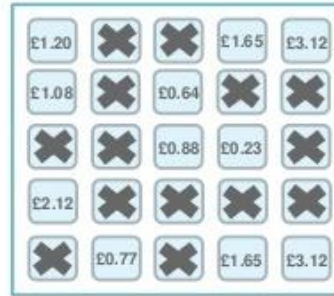
Today **Real Time Bidding (RTB)** is the industry standard in online ad pricing.

Each ad-space or SEA # position is valued/priced according to its actual through and demand-supply equilibrium (via bidding)

*Tip: Do not be confused about CPM-CPC-CPA and RTB. The first are the models where players decide the basis of pricing (impressions-click or conversion)
No matter what price basis is chosen > Next Step: most ad (spaces) are being priced on the basis of an auction / real time bidding.
For example: Ad space is offered based on 1000 thousand impressions (CPM) Then all brands put a bid to get this CPM based ad space.*



Vs.



Purchasing in bulk even though each impression has different value to campaign

Purchase impressions based on their individual worth and only those which are valuable

Real-Time Bidding: Second-Price Auction

Google uses variants of so-called “**second-price auction**” The buyer does not have to pay highest bid but only that of the **next highest bidder**
But why does Google use second price auction (bidding) which is seemingly more complicated?

Why wouldn't Google simply want to get the maximum bid offered by the highest bidder? The answer: Collaborative Game Theory

First-Price Auction



Second-Price Auction



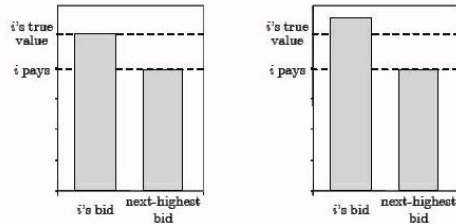
Real-Time Bidding: Second-Price Auction

The answer is to **prevent over/underbidding** and to find the **real value** of each ad space (Game-Theoretic Proof)

Let's say: **Firm A bids too high** just to "win" – and the other firms are realistic/honest about the actual value – **Firm A pays too high.**

Let's say: **Firm A bids too low** – assuming that others will also keep their bids (and lower than Firm A) – most likely **Firm A never wins.**

Thus: In second-price auction the bidders rely on their (collective) honesty on the actual value of ad space.



- Bidding honestly, i is the winner
- If i bids higher, he will still win and still pay the same amount

How a second-price auction works:

*All advertisers bid and the winner pays \$0.01 more than the second highest bidder



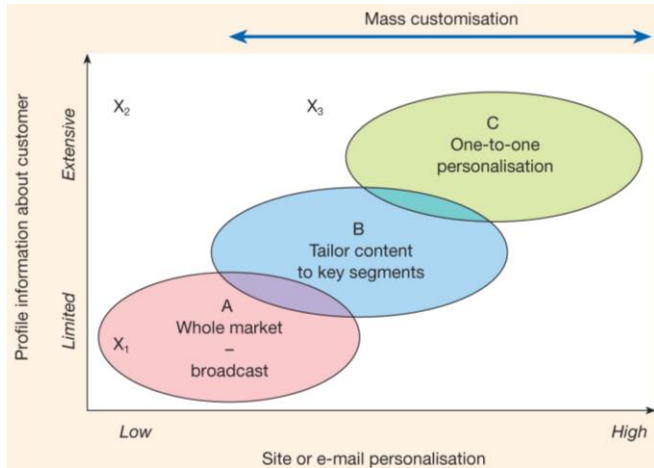
Personalized Targeting & Advertising

Retargeting, Contextual and Behavioral Targeting

Personalization Levels

Firms make decisions on whether: they target **all customer base** with same message (mass – which is no longer common nowadays) or:

- 1 **Segments:** targeting to certain segments (with multiple messages) – less cost than individualization
- 2 **One-to-one:** individualized (personalized) targeting for each customer: possible through big-data & analytics



TARGETING TO SEGMENTS



ONE-TO-ONE PERSONALIZATION



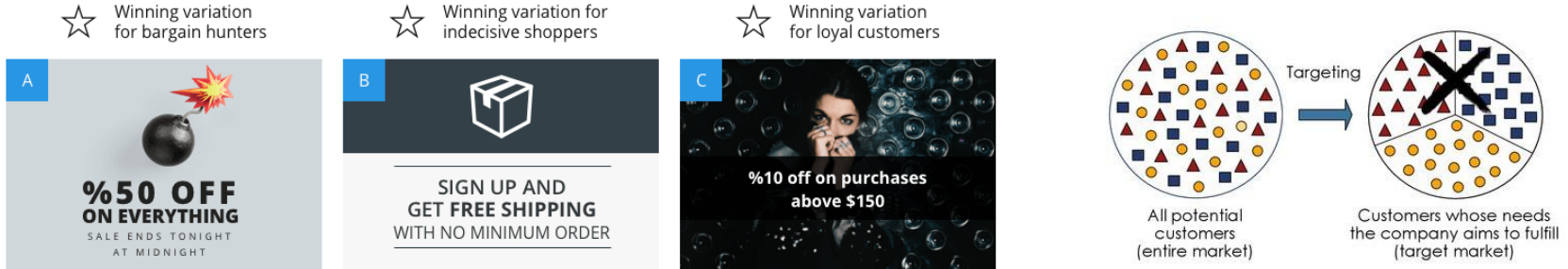
Personalization Levels: Segmentation

Not fully individualized personalization: different offers, limited in number, are tailored to certain number of segments

The messages (that user see) **are not fully tailored for the user**.

The user (and the other users in the same segment) – receive the same messages, whereas other segments receive different ads.

Example: Here we have three different screens for (i) bargain hunters (ii) indecisives and (iii) loyal customers. Not everybody gets a special treatment like in full-individualization. But the campaign/ad still takes their potential preferences into consideration.



A/B Testing – Helps in Segment & Individual Based Targeting

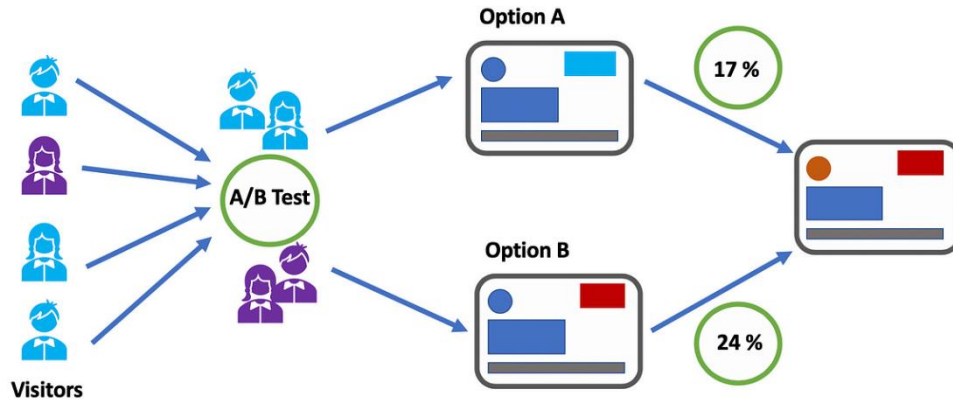
- A/B testing is a method that is commonly used by **marketers** and researchers to **compare two versions** of a particular variable in a given campaign.
- It splits a group of people into **two random groups- group A and group B** - and exposing them to different versions of the variable that is being tested.
- Then, statistical analysis is used to compare the performance of the two groups.



A/B Testing – How Does it Work?

In everyday life, many factors can influence whether someone clicks on something, (i.e device, size of the button etc). This is where **randomization** can be useful.

- **Randomization** helps to reduce the impact of other factors, like using a **mobile or desktop** device or having different **button sizes**, which might influence the results of the experiment.
- By randomly assigning people to the different groups, the results are more **representative of the entire population**

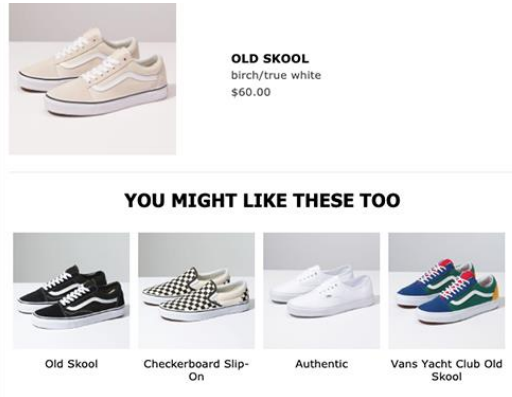


Personalization Levels: Individual-Level (Full) Personalization

Users get individualized (tailored) offers, ads and communication messages based on their:

- Purchase, Search, Browse history 购买、搜索、浏览历史记录
- Their likes, hobbies, interests and lifestyles • 用户的喜好、爱好、兴趣和生活方式
- Location, demographic and life-cycle characteristics • 地理位置、人口统计特征和生命周期特征

Food for Thought: Think of your shopping. In today's digital world you get personalized offers targeted on the basis of your individual preferences. However full-individualization is used only when it is necessary and has real potential to pay off. Above all it is more costly, and requires more resources (time-people) than segmentation.

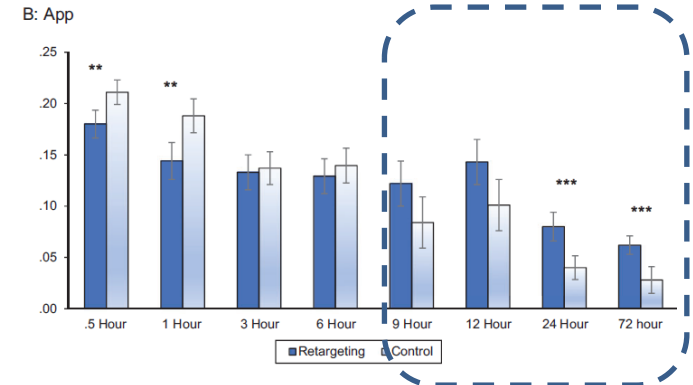
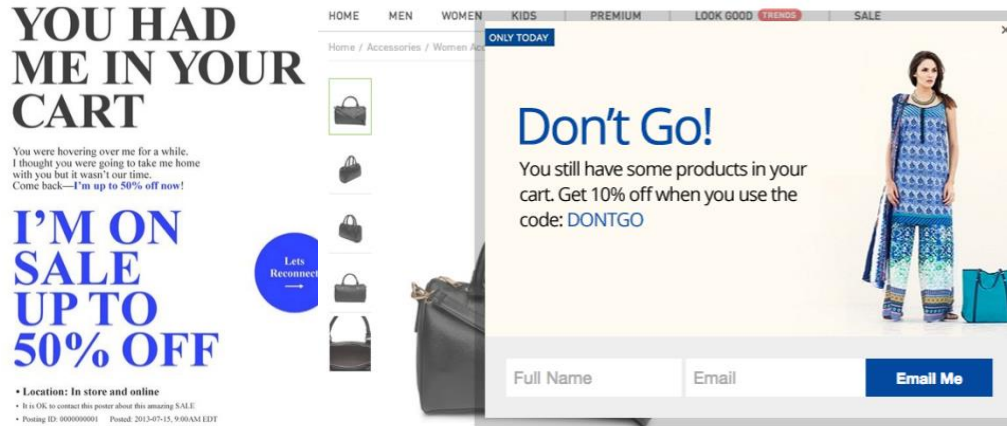


Targeting in New Forms: Shopping-Cart Retargeting:

Consumers often abandon e-commerce carts: companies shift their online advertising budgets to immediate: **E-Cart Retargeting (ECR)**.

However this may backfire: Customers who received an early ECR ad within **30 minutes to one hour** after cart abandonment are **less likely to make a purchase**.

In other words: **too-early ECR ads can engender worse purchase rates** than without delivering them, thus wasting online advertising budgets.



* DARK blue bars represent the customers who receive EC retargeting – where LIGHT blue represents who don't.

Personalized Targeting: Types of Individualization

- 1 Contextual:** KLM ad after you visit a travel website/ or travel blog: Content of your landing page is the basis)
- 2 Retargeting:** Zalando ad after visiting Zalando website: you get the ad at some point after you visit brand's webpage
- 3 Behavioral :** Car insurance ad after searching to buy a new car: Your search-keyword/queries are the basis for personalization
- 4 Geographical:** Ajax ad only for Amsterdam area (or push notification of a mall in the proximity of that mall)

Dynamic (Multidimensional) Targeting Combines and real-time uses: search + previous visit/search on different website (factors given above)



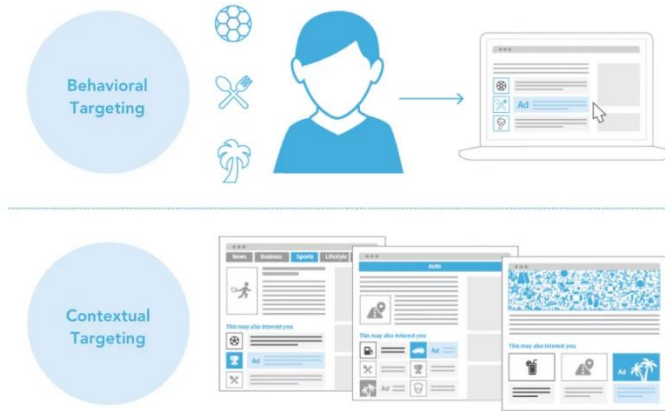
Personalized Targeting: Types of Individualization

Check the difference between Contextual and Behavioral targeting:

Behavioral: mainly based on recent search & click behavior. Search queries and clicks (visits) is the base of targeting.

Contextual: based on the context of the websites visited. The content (thema, tags, keywords) on the visited website is the basis of targeting.

It is also possible that both keywords and website contents can be used jointly in contextual targeting.



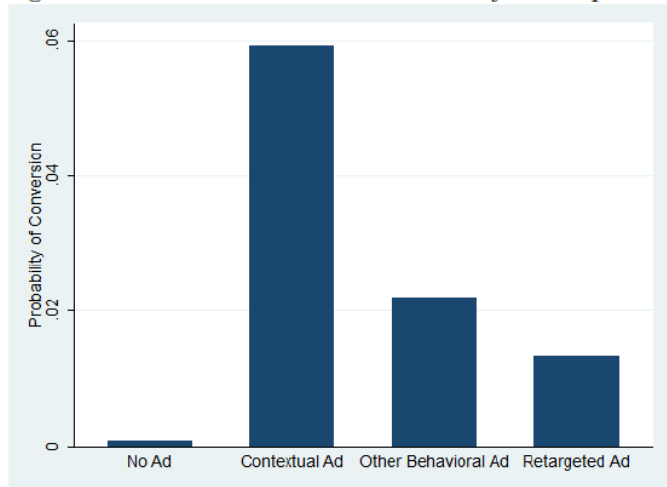
Personalized Targeting Methods: Efficiency & Comparison

Research shows that among different types, **CONTEXTUAL** personalized targeting is the most effective at same-day ad exposure.

RETARGETING is found to be less effective. Can you tell why?

In Retargeting those people are targeted who visited the same website before – but doesn't that also mean they did not find what they were looking for.

Figure 2: Conversion Rate with Same-Day Ad Exposure



But is Retargeting always less effective?

Let's see the next slide.

Positive Emotions in Search: Impact on Clicks & Engagement

Research on how **positive incidental emotions** (i.e. good mood not caused by the search itself) influence **search behavior and clicking on paid ads**

They find: Even though those emotions are **irrelevant** to the task, they “spill over” to affect keyword choice and ad click-through rates.

When a search query contains positive emotion words, users are less likely to activate persuasion knowledge (i.e. less skeptical of ads), which in turn increases click probability on paid search results.

Table 1. List of Queries with Positive Emotion, Nonemotion, and No Descriptive Keywords.

Queries with Positive Emotion Keywords	Queries with Nonemotion Keywords	Queries with No Descriptive Keywords
Romantic hotels ^a	International hotels	Hotels
Sexy sundresses ^a	Pink sundresses	Sundresses
Lovely flowers ^b	Red flowers	Flowers
Calming foods ^c	International foods	Foods
Energetic art ^d	Animated art	Art
Delightful fruit ^e	Juicy fruit	Fruit
Playful kitten ^e	Newborn kitten	Kitten
Sentimental jewelry ^b	Customized jewelry	Jewelry
Exciting comics ^d	Colorful comics	Comics
Love poems ^b	Short poems	Poems
Upbeat jazz songs ^f	Contemporary jazz songs	Jazz songs

Notes: Emotion descriptors correspond to Richins's (1997) positive emotion categories: ^aromantic love, ^blove, ^cpeacefulness, ^dexcitement, ^ejoy, and ^foptimism.

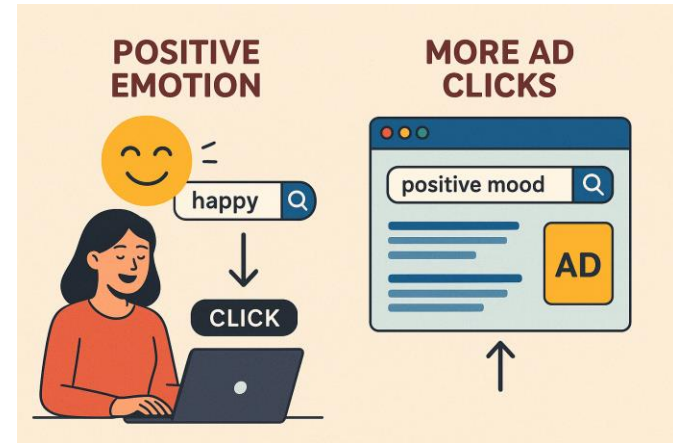
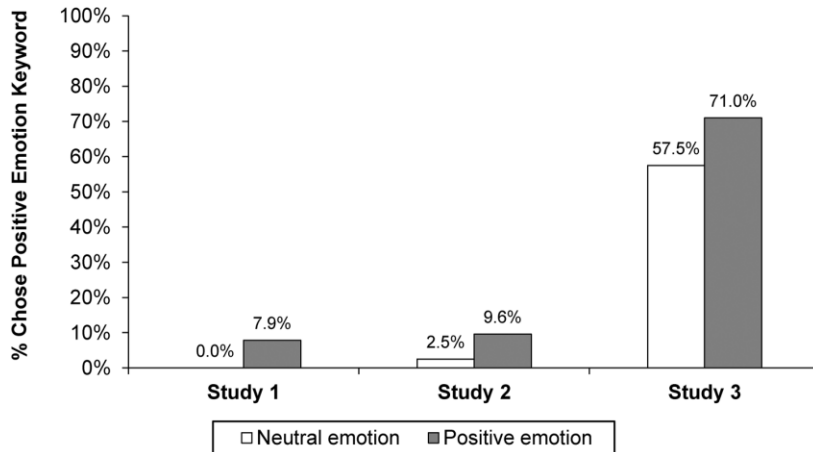


Positive Emotions in Search: Impact on Clicks & Engagement

Marketers should consider adding **positive emotion words** (e.g. “happy,” “joyful”) in their paid search bidding or targeting, not just product attribute keywords.

It may pay off to focus on times or contexts when consumers are more likely to be in a **positive mood** (weekends, holidays).

Hedonic / emotional products (clothing, entertainment, gifts) may especially benefit from positive emotion keyword targeting.



Thank You